

Mechanics of Materials

Equations

$$\sigma = \frac{\mathbf{P}}{A} \quad \sigma_B = \frac{\mathbf{P}_B}{A_B} \quad \delta = \varepsilon L$$

$$\delta = \frac{\mathbf{P}L}{EA} \quad \delta_T = \alpha \Delta T L$$

$$\delta = \sum_{i=1}^n \frac{\mathbf{N}_i L_i}{E_i A_i} \quad \delta = \int_0^L \frac{\mathbf{N}(x)}{EA(x)} dx$$

$$\text{Elongation} = \frac{L_1 - L_0}{L_0} 100\%$$

$$\text{Area Reduction} = \frac{A_0 - A_1}{A_0} 100\%$$

$$E = \frac{\sigma}{\varepsilon} \quad \nu = -\frac{\varepsilon'}{\varepsilon} \quad G = \frac{E}{2(1+\nu)}$$

$$\tau_{\text{avg}} = \frac{\mathbf{V}}{A} \quad \tau = G\gamma(\text{radians})$$

$$n = \frac{\text{actual strength}}{\text{required strength}} = \frac{\text{yield stress}}{\text{allowable stress}}$$

$$\mathbf{P} = k\delta \quad k = \frac{\mathbf{P}}{\delta} = \frac{1}{f} \quad f = \frac{\delta}{\mathbf{P}} = \frac{1}{k}$$

$$\varepsilon_T = \alpha \Delta T \quad \delta_T = \alpha \Delta T L \quad \sigma_T = E \alpha \Delta T$$

$$\sigma_{\text{max}} = \sigma_x \quad \tau_{\text{max}} = \frac{\sigma_x}{2}$$

$$\tau_{\text{single}} = \frac{\mathbf{V}}{A_{\text{pin}}} \quad \tau_{\text{double}} = \frac{\mathbf{V}}{2A_{\text{pin}}}$$

$$\sigma_{B, \text{ single}} = \frac{\mathbf{P}}{d_{\text{pin}} t_{\text{bracket}}} \quad \sigma_{B, \text{ double}} = \frac{\mathbf{P}}{2d_{\text{pin}} t_{\text{bracket}}}$$

$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\sigma_{y1} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\sigma_{x1} + \sigma_{y1} = \sigma_x + \sigma_y \quad \tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{1,2} = \text{center} \pm \text{radius}$$

$$\theta_s = \theta_p \pm 45^\circ \quad \sigma_1 \geq \sigma_2 \geq \sigma_3 \quad \theta = \frac{\phi}{L}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \theta = \frac{d\phi}{dx} \quad k_T = \frac{T}{\phi}$$

$$\gamma_{\text{max}} = r \frac{d\phi}{dx} = r\theta \quad (\gamma \text{ is measured in radians})$$

$$\gamma_\rho = \frac{\rho}{r} \gamma_{\text{max}} \quad \tau = Gr\theta = \frac{\rho}{r} \tau_{\text{max}}$$

$$I_p = \frac{\pi}{32} (d_2^4 - d_1^4) \quad \tau = \frac{16T}{\pi d^3} \text{ (solid only)}$$

$$\tau_{\text{max}} = \frac{T r}{I_p} \quad \tau = \int_0^L \frac{T(x)}{I_p(x)} dx$$

$$\phi = \frac{TL}{GI_p} \quad \phi = \int_0^L \frac{T(x)}{GI_p(x)} dx$$

$$\varepsilon_{\text{max}} = \frac{\gamma}{2} \quad \gamma = \frac{\tau}{G} \quad G = \frac{E}{2(1+\nu)}$$

$$|\sigma_C| = |\sigma_T| = |\tau_{\text{max}}| \quad (\text{in pure shear only})$$

$$\psi = \Delta\theta \quad \omega = T\psi \quad \omega = 2\pi f$$

$$P = \frac{d\omega}{dt} = T \frac{d\psi}{dt} = T\omega$$

$$P = 2\pi f T \quad P = \frac{2\pi n T}{60}$$

$$\text{RPM} : n = 60f \quad H(\text{Hp}) = \frac{2\pi n T}{33,000} \quad \left(\text{Hp} = \frac{\text{ft} \cdot \text{lb}}{\text{s}} \right)$$

$$T = 2f A_m \quad \tau = \frac{T}{2t A_m} \quad u = \frac{T^2 L}{2GJ}$$

$$T_1 = T \frac{G_1 I_{p1}}{G_1 I_{p1} + G_2 I_{p2}} \quad T_2 = T \frac{G_2 I_{p2}}{G_2 I_{p2} + G_1 I_{p1}}$$

$$J = \frac{4tA_m^2}{L_m} \quad \phi = \frac{TL}{GJ} \quad V = \frac{dM}{dx}$$

$$\kappa = \frac{1}{\rho} \quad \sigma = -\frac{y}{\rho} = -\kappa y \quad \kappa = \frac{M}{EI}$$

$$\delta = \rho(1 - \cos\theta) \quad \sin\theta = \frac{L/2}{\rho}$$

$$\sigma_x = -\frac{Ey}{\rho} = -E\kappa y \quad \sigma_x = -\frac{My}{I} = \frac{M}{S}$$

$$\sigma = \frac{P}{A} - \frac{My}{I} \quad \tau = \frac{VQ}{Ib} \quad S = \frac{I}{y}$$

$$\sigma = \frac{P}{A} + \frac{Pey}{I} \quad y_0(\text{adjusted NA}) = -\frac{1}{Ae}$$

$$Q = (\text{Area})(\text{lever arm from NA to centroid})$$

$$\tau_{\max}(\text{rect.}) = \frac{3V}{2A} \quad \tau_{\max}(\text{solid}) = \frac{4V}{3A}$$

$$\tau_{\max}(\text{hollow}) = \frac{4V}{3A} \left(\frac{(r_1 + r_2)^2}{r_1^2 + r_2^2} \right)$$

$$V_{\text{web}}(\text{wide flange}) = \frac{bh_1}{3}(2\tau_{\max} + \tau_{\min})$$

$$I = \sum_{i=1}^N (I_i + A_i d_i^2) \quad d_i = d_{cm} - d_{NA}$$

$$M = \kappa(E_1 I_1 + E_2 I_2) \quad \bar{y} = \frac{\Sigma y dA}{\Sigma dA}$$

$$\sigma_{x1,y1} = -n \frac{My}{I_T} \quad f = \frac{VQ}{I}$$

$$\sigma(\text{spherical}) = \frac{pr^2}{2r_mt} = \frac{pr}{2t}$$

$$\sigma_1(\text{hoop}) = \frac{pr}{t} \quad \sigma_2(\text{axial}) = \frac{pr}{2t}$$

$$\tau_{\max}(\text{in plane}) = \frac{pr}{4t} \quad \tau_{\max}(\text{any plane}) = \frac{pr}{2t}$$

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2} \right)^2 + \left(\frac{\gamma_{xy}}{2} \right)^2}$$

$$\varepsilon_{x'} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2} \cos(2\theta) + \frac{\gamma_{xy}}{2} \sin(2\theta)$$

$$\varepsilon_{y'} = \frac{\varepsilon_x + \varepsilon_y}{2} - \frac{\varepsilon_x - \varepsilon_y}{2} \cos(2\theta) - \frac{\gamma_{xy}}{2} \sin(2\theta)$$

$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} \quad \tan(2\theta_s) = -\frac{\varepsilon_x - \varepsilon_y}{\gamma_{xy}}$$

$$S_{\text{req}} = \frac{M_{\max}}{\sigma_{\text{allow}}} \quad EI \frac{d^2 v}{dx^2} = M(x) \quad F = qs$$

$$\sigma_{cr} = \frac{\pi^2 E}{\left(\frac{KL}{r} \right)^2} \quad P_{cr} = \frac{\pi^2 EI}{(KL)^2} \quad \frac{1}{\rho} = \frac{M}{EI}$$

Column Boundary Conditions

- Pinned-Pinned K=1
- Fixed-Fixed K=0.5
- Pinned-Fixed K=0.7
- Fixed middle, free ends K=2

Releases

- Axial release

$$N = 0, \quad V, M \neq 0$$

- Shear release

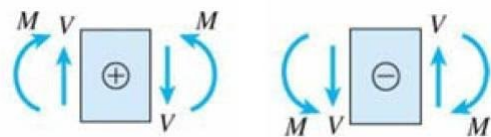
$$V = 0, \quad N, M \neq 0$$

- Moment release

$$M = 0, \quad N, V \neq 0$$

- Torque release

$$T = 0, \quad N, V, M \neq 0$$



Deformation sign convention